

Exploratory Data Analysis (EDA)

Dataset: EDA - EDA.csv

1. Aim

To perform Exploratory Data Analysis on the given dataset to understand its structure, variables, missing values, duplicate records, measurement types, and availability of variable descriptions using Python.

2. Dataset Overview

Property	Result
Number of rows	28
Number of columns	4
Missing values in Description	6
Missing values in Measurement	12
Duplicate rows	0

3. Python Code

```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("EDA - EDA.csv")

# Basic information
print(df.head())
print("Shape:", df.shape)
print(df.info())
print(df.describe(include="all"))

# Missing values
print("\nMissing Values:")
print(df.isnull().sum())

# Duplicate values
print("\nDuplicates:", df.duplicated().sum())

# Measurement type analysis
measurement_count = df["Measurement"].fillna("Missing").value_counts()
measurement_count.plot(kind="bar")
plt.title("Variables by Measurement Type")
plt.xlabel("Measurement Type")
plt.ylabel("Number of Variables")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# Description availability
description_status = df["Description"].notna().map(
    {True: "Available", False: "Missing"}
).value_counts()

description_status.plot(kind="bar")
plt.title("Description Availability")
plt.xlabel("Description Status")
plt.ylabel("Number of Variables")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

4. Graphs

The following graphs summarize the structure of the supplied data dictionary.

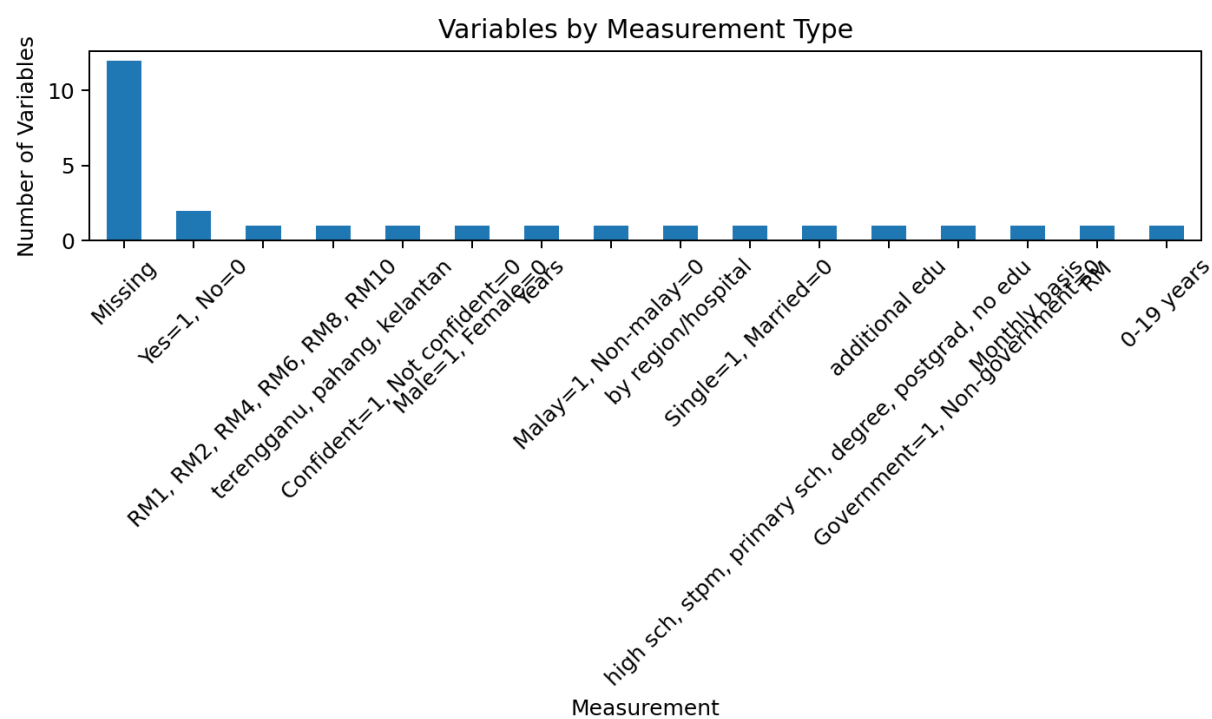


Figure 1: Distribution of variables by measurement type.

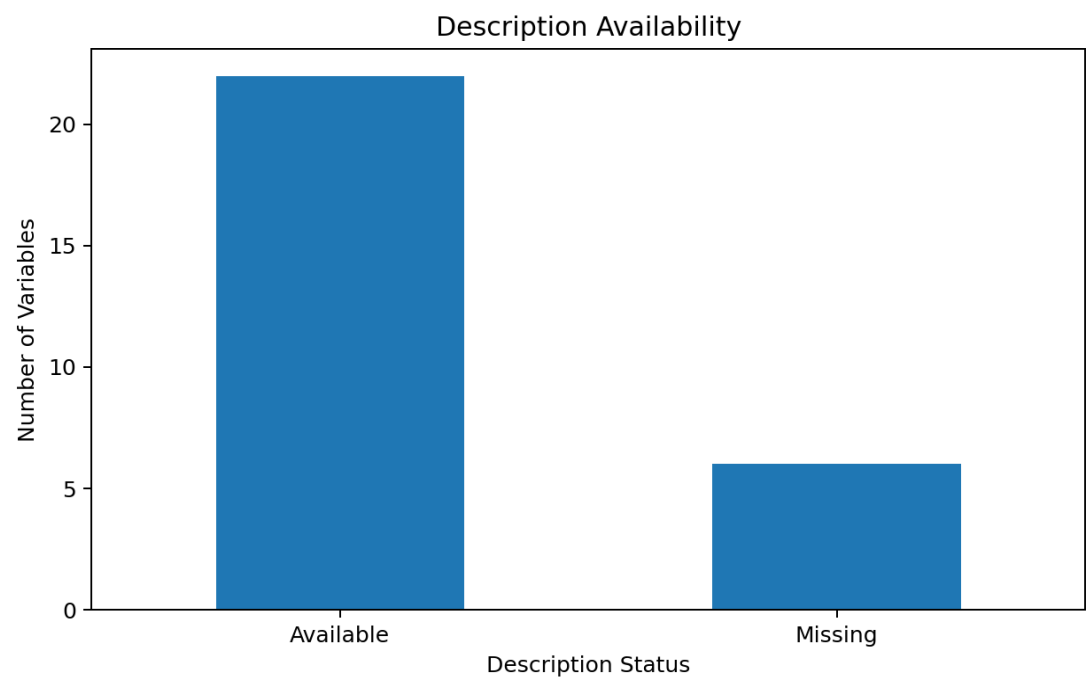


Figure 2: Availability of descriptions for the variables.

5. Observations

The dataset contains 28 variables/records and 4 columns. There are 0 duplicate rows. The Description column has 6 missing entries, while the Measurement column has 12 missing entries. The graphs provide a visual summary of measurement information and description availability.

6. Conclusion

EDA was successfully performed on the given data dictionary. The analysis helped identify the structure of the dataset, missing information, duplicate records, and the distribution of measurement types. The results can be used as a preliminary understanding of the variables before further analysis.